Scrambled states survive local Pauli noise up to the hashing entropy

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Information is encoded in a random pure state from a state 2-design, such as a random stabilizer state, and one layer of Pauli noise then acts with no further randomization. We bound the optimal probability of telling two such encoded states apart at every system size, using the information spectrum of the Pauli errors. For i.i.d. noise the states stay distinguishable until the error entropy reaches one bit per qubit, then become indistinguishable through a Gaussian crossover. Purity-based bounds used for error mitigation give only a weaker threshold.

Introduction.—A common model of a quantum computation that scrambles its input is a random state: the encoding is drawn from a state 2-design, an ensemble whose first two moments match Haar-random states, such as random stabilizer states [1]. Hardware noise then acts on the encoded state. How much of the encoded information survives? We ask this in its simplest operational form. Two pure states ψ_1, ψ_2 are drawn independently from a state 2-design. Both pass through the same Pauli channel $\mathcal{N}$, and nothing acts afterward. An observer who knows the two candidates measures the output to decide which one was prepared. With equal priors, the best success probability is $(1+\overline{T})/2$, where

$$\overline{T} = \mathbb{E}_{\psi_1,\psi_2} \frac{1}{2} \|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1. \tag{1}$$

The order of operations matters. The 2-design describes how the encoded states are chosen; it does not act on the noise. If another 2-design unitary U followed the noise, the effective channel $U^\dagger \mathcal{N}(U \cdot U^\dagger)U$ averaged over U is global depolarization $\rho \mapsto \theta\rho + (1-\theta)I/d$, with $\theta = (d^2 p_I - 1)/(d^2-1)$ [2]. For local depolarizing noise $p_I = (1-3r/4)^n$, so θ decays exponentially in n for every $r > 0$ and there is no threshold. Without the trailing randomization, the observer can exploit the local structure of the noise. At $n = 8$ and $r = 0.2$, for example, the Haar-averaged distance is 0.736 (Fig. 1), versus about 0.272 after twirling.

This question underlies known costs of quantum error mitigation (QEM). Takagi, Tajima, and Gu [3] and Tsubouchi, Sagawa, and Yoshioka [4] showed that recovering ideal expectation values from noisy samples needs a number of samples that grows exponentially with circuit depth; see also Refs. [5, 6]. Their bounds concern a worst-case pair of inputs and contract a relative entropy toward I/d at every layer, so they predict exponential cost for any noise strength. Quek *et al.* [7] showed that random circuits make this cost appear at very shallow depth. Their mechanism is the one studied here: a 2-design spreads the state over high-weight Pauli operators, which local depolarizing noise of strength r damps by $(1-r)^w$ at weight w. They control the outcome through the purity, which certifies decay only for $(1-r)^2 < 1/3$, i.e., $r > 0.423$.

For the pairwise quantity (1), Ibarrondo and Stilck França [8] proved that the average contraction tends to one when the Choi-state entropy satisfies $S(\tau_\mathcal{N}) < -\log_2 \|\mathcal{N}(I/d_{\text{in}})\|_\infty$; for Pauli channels this means the error entropy is below n bits. Combined with a 2-design upper bound, this gives local-depolarizing thresholds near 0.25 and 0.42, leaving the interval between them open. Liu *et al.* [9] found the same two values by decoupling, with an $O(n^{-1/2})$ upper bound in between. Sommers *et al.* [10] identified the hashing transition and an $n^{-1/2}$ window in the coherent information of random codes.

Here we give two-sided bounds on $\overline{T}$ that hold for every Pauli distribution and system size and are controlled by a single quantity, the information spectrum of the error distribution. For i.i.d. noise they locate the threshold exactly at the hashing entropy and resolve the crossover as a Gaussian profile. The method is narrow and combines standard ingredients: the lower bound is the typical-image projector of Ref. [8] applied to Pauli errors, and the upper bound is a second-moment estimate of the kind used in privacy amplification and quantum soft covering [11–13]. We also determine which conclusions depend only on the design's second moments.

Setup.—Let $d = 2^n$, $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, and

$$\mathcal{N}_p(\rho) = \sum_{P \in \mathcal{P}_n} p_P P\rho P, \qquad s(P) = -\log_2 p_P, \tag{2}$$

where the surprisal s is a random variable with $P \sim p$. The distribution p is arbitrary, so errors may be correlated across qubits. Logarithms are base two. The channel does not depend on the state samples. Each $\psi_i = |\psi_i\rangle\langle\psi_i|$ is drawn independently from an exact pure-state 2-design on the full d-dimensional space:

$$\mathbb{E}\psi_i = I/d, \qquad \mathbb{E}\psi_i^{\otimes 2} = (I+F)/[d(d+1)], \tag{3}$$

where F swaps the two copies. The two ensembles need not coincide.

Theorem 1. *For every Pauli distribution p and $a \geq 0$,*

$$\mathbb{P}[s < n-a] - 2^{-a} \leq \overline{T} \leq \mathbb{P}[s < n+a] + 2^{-a/2}/\sqrt{2}. \tag{4}$$

More generally, for any subset $\mathcal{G} \subseteq \mathcal{P}_n$ and any partition

$\{\mathcal{C}_j\}$ of $\mathcal{P}_n$, with $m_j = \sum_{\mathcal{C}_j} p_P$ and $v_j = \sum_{\mathcal{C}_j} p_P^2$,

$$\overline{T} \geq p(\mathcal{G}) - |\mathcal{G}|/d, \tag{5}$$

$$\overline{T} \leq \sum_j \min\left\{ m_j, \sqrt{\frac{d^2 v_j - m_j^2}{2(d+1)}} \right\}. \tag{6}$$

Proof. Fix ψ_1 and let Π project onto the span of $\{P|\psi_1\rangle : P \in \mathcal{G}\}$. For each realization, the variational formula gives $\frac{1}{2}\|\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)\|_1 \geq \operatorname{tr}\Pi[\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)]$. The first term is at least $p(\mathcal{G})$. Averaging the second over the independent ψ_2 uses only $\mathbb{E}\psi_2 = I/d$ and unitality, giving $\operatorname{tr}\Pi/d \leq |\mathcal{G}|/d$. Averaging over ψ_1 proves (5); only the second state needs to be a 1-design. This is the typical-Kraus-image argument of Ref. [8] (Theorem 18 there), with an arbitrary set of Pauli errors in place of a typical set.

For the upper bound let $\Delta = \psi_1 - \psi_2$ and $\mathcal{N}_j(\rho) = \sum_{\mathcal{C}_j} p_P P\rho P$. Pauli orthogonality and (3) give

$$\mathbb{E}\operatorname{tr}\mathcal{N}_j(\psi_i)^2 = \frac{dv_j + m_j^2}{d+1}, \qquad \mathbb{E}\operatorname{tr}\mathcal{N}_j(\psi_1)\mathcal{N}_j(\psi_2) = \frac{m_j^2}{d},$$
$$\mathbb{E}\operatorname{tr}\mathcal{N}_j(\Delta)^2 = \frac{2d}{d+1}v_j - \frac{2m_j^2}{d(d+1)}. \tag{7}$$

Jensen's inequality and $\|A\|_1 \leq \sqrt{d}\|A\|_2$ give the square-root term in (6); the triangle inequality gives $\frac{1}{2}\|\mathcal{N}_j(\Delta)\|_1 \leq m_j$. Sum over classes.

For (4), take $\mathcal{G} = \{s < n - a\}$, which has fewer than $d2^{-a}$ elements, and split the partition at $s = n+a$: bound the head by its mass and the tail by $\sqrt{dv/2}$ with $v \leq 2^{-n-a}$. □

The two cumulative probabilities in (4) are evaluated at different points, so the result is a bracket rather than an identity with $\mathbb{P}[s < n]$. It is informative when the surprisal concentrates. The event $s < n$ means $p_P > 1/d$; the test detects a set of error images, but does not identify individual errors or recover the state. The lower bound vanishes unless some error has $p_P > 1/d$, which for i.i.d. noise requires $\max_i p_i > 1/2$, i.e., $r < 2/3$ for depolarizing noise.

The single-class partition, $\mathcal{C} = \mathcal{P}_n$, reduces (6) to the purity of the noisy output, $\overline{T} \leq [d\sum_P p_P^2/2]^{1/2}$. This is the quantity that controls the error-mitigation bounds of Ref. [7]. For local depolarizing noise $\sum_P p_P^2 = [(1 + 3(1-r)^2)/4]^n$, which decays faster than d^{-1} only for $r > 1 - 1/\sqrt{3} = 0.423$. Splitting off the likely errors is what lets the bracket follow the distance below that value.

Hashing threshold for i.i.d. noise.—Let $p = \boldsymbol{p}^{\otimes n}$ with single-qubit distribution $\boldsymbol{p} = (p_0, p_x, p_y, p_z)$, entropy $H = \mathbb{E}[-\log_2 p_i]$, and variance $\sigma^2 = \operatorname{Var}[-\log_2 p_i]$. The surprisal is then a sum of n i.i.d. terms with mean nH. For $H < 1$, choosing $a = \epsilon n$ with $0 < \epsilon < 1 - H$ and a large-deviation bound [14] give $\overline{T} \geq 1 - e^{-\Omega(n)}$. For $H > 1$ the upper bound gives $\overline{T} \leq e^{-\Omega(n)}$. The threshold $H = 1$ is the zero of the coherent information $1 - H$ of the Pauli channel at maximally mixed input [15, 16]. It is the threshold for this discrimination task, not a general quantum-capacity boundary: degenerate codes transmit quantum information beyond it [17]. For depolarizing noise $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, with $q = 3r/4$ and $\boldsymbol{p} = (1-q, q/3, q/3, q/3)$, the threshold is $q_c = 0.189290\ldots$, or $r_c = 0.252386\ldots$. Thus local depolarizing noise is tolerated up to r_c, while the purity bound above certifies loss only beyond 0.423.

The decay above r_c also follows from existing results, so we do not present it as new. The unsmoothed Rényi decoupling bound of Ref. [12] applies because, for any reference-system operators X_P, Eq. (3) gives

$$\mathbb{E}_\psi \Big\| \sum_P (P\psi P - I/d) \otimes X_P \Big\|_2^2 = \frac{d}{d+1}\sum_P \|X_P\|_2^2 - \frac{\|\sum_P X_P\|_2^2}{d(d+1)} \leq \|X\|_2^2. \tag{8}$$

Theorem 8 of Ref. [12] then gives, for $1 < \alpha \leq 2$,

$$\mathbb{E}\|\mathcal{N}_p(\psi) - I/d\|_1 \leq 2^{2/\alpha - 1} 2^{-\frac{\alpha-1}{\alpha}[H_\alpha(p) - n]}, \tag{9}$$

which decays once α is close enough to one that $H_\alpha(\boldsymbol{p}) > 1$. The smoothed bound of Ref. [9] gives the same conclusion with smoothing parameter $2^{-\kappa n}$ for small $\kappa > 0$. (Its Lemma 1 derives from the min-entropy AEP [18] through $H_{\min} \leq H_2$, Eq. (48) of Ref. [19] at $\alpha \to \infty$; the two references use different smoothing conventions, which the unsmoothed route avoids.)

What the bracket adds is the crossover. Let the single-qubit distribution $\boldsymbol{p}_n$ depend on n, with i.i.d. errors at each n, and set $x_n = \sqrt{n}(1 - H_n)/\sigma_n$.

Corollary 1. *If $\boldsymbol{p}_n \to \boldsymbol{p}_*$ with $H(\boldsymbol{p}_*) = 1$ and $\sigma(\boldsymbol{p}_*) > 0$, then uniformly over independent exact 2-design inputs,*

$$\overline{T}_n = \Phi(x_n) + O\left(\frac{\log n}{\sqrt{n}}\right), \tag{10}$$

where Φ is the standard normal distribution function.

Proof. With $\rho_n = \mathbb{E}|-\log_2 p_{n,i} - H_n|^3$ and $b_n = C_{\mathrm{BE}}\rho_n/(\sigma_n^3\sqrt{n})$, the Berry–Esseen theorem [20] applied to (4) gives

$$\overline{T}_n \geq \Phi\left(x_n - \frac{a}{\sqrt{n}\sigma_n}\right) - b_n - 2^{-a},$$
$$\overline{T}_n \leq \Phi\left(x_n + \frac{a}{\sqrt{n}\sigma_n}\right) + b_n + 2^{-a/2}/\sqrt{2}. \tag{11}$$

On a four-letter alphabet the third moment is bounded uniformly, since $t|\log t|^3 \to 0$ at zero, and σ_n stays bounded away from zero. Hence $b_n = O(n^{-1/2})$; set $a = 2\log_2 n$. □

A transverse crossing of $H = 1$ therefore has a noise window of width $n^{-1/2}$ with the universal profile Φ. Figure 1 shows the finite-size bracket and the approach to Φ.

Correlated noise.—The bracket holds for any Pauli distribution, but the Shannon entropy need not then locate the threshold. Global depolarization $\mathcal{N}(\rho) = \theta\rho + (1-\theta)I/d$, $0 < \theta < 1$, has $p_I = \theta + (1-\theta)/d^2$ and $p_P = (1-\theta)/d^2$ otherwise. Its entropy rate tends to $2(1-\theta)$, above one for $\theta < 1/2$, yet $\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2) = \theta(\psi_1 - \psi_2)$ gives $\overline{T} \to \theta$. The bracket recovers this limit: with $a = n/2 - 1$, only the identity has surprisal below $n - a$, every other error lies above $n + a$, and $|\overline{T} - \theta| \le 2^{-n/4+1}$ for large n.

What the second moments do not fix.—The bounds above hold for every exact 2-design. Two further statements do not, which limits conclusions stated for an arbitrary design of a given order [8, 9].

Degenerate critical noise. The variance condition in Corollary 1 cannot be dropped. Complete dephasing, $\boldsymbol{p}_* = (1/2, 0, 0, 1/2)$, has $H = 1$ and $\sigma = 0$. In dimension $d = 2^n$ the uniform union of $d + 1$ mutually unbiased bases (MUBs) is an exact 2-design [21]; take the computational basis as one of them. Its d states are unchanged by dephasing, and every other state maps to I/d. Counting pairs,

$$\overline{T}_{\mathrm{MUB}} = \frac{(d-1)(2d+1)}{d(d+1)^2} \longrightarrow 0. \tag{12}$$

For Haar states the dephased outputs are diagonal, so their trace distance is the total-variation distance between two independent Dirichlet$(1, \dots, 1)$ vectors. Single coordinates $X, Y \sim \mathrm{Beta}(1, d-1)$ satisfy $\mathbb{E}\min(X, Y) = \int_0^1 (1-t)^{2d-2} dt = (2d-1)^{-1}$, hence

$$\overline{T}_{\mathrm{Haar}} = \frac{d}{2}\mathbb{E}|X - Y| = \frac{d-1}{2d-1} \longrightarrow \frac{1}{2}. \tag{13}$$

The two ensembles share their first two moments, yet their limits differ (Fig. 2). This strengthens the observation of Ref. [8] that computational-basis states, a 1-design, are not contracted by dephasing. Positive variance at each n is also not enough. Set $\epsilon_n = n^{-6}$ and $\boldsymbol{p}_n = (1/2 - \epsilon_n + \delta_n, \epsilon_n, \epsilon_n, 1/2 - \epsilon_n - \delta_n)$, with $\delta_n > 0$ solving $H(\boldsymbol{p}_n) = 1$. Binary entropy bounds give $\delta_n = O(\sqrt{\epsilon_n \log(1/\epsilon_n)})$, so $n\|\boldsymbol{p}_n - \boldsymbol{p}_*\|_1 \to 0$. Replacing one tensor factor at a time and using $\|\mathcal{A}\otimes\mathcal{B} - \mathcal{A}'\otimes\mathcal{B}'\|_\diamond \le \|\mathcal{A} - \mathcal{A}'\|_\diamond + \|\mathcal{B} - \mathcal{B}'\|_\diamond$, the n-qubit channels differ in diamond norm by at most this quantity. MUB inputs still give $\overline{T}_n \to 0$, although $\sigma_n > 0$ and $x_n = 0$ for every n.

Decay rates. For i.i.d. errors, grouping Pauli strings into empirical types t in (6) gives $\overline{T}_n \le \mathrm{poly}(n) 2^{-nE(\boldsymbol{p})}$, using standard type bounds [14], with

$$\begin{aligned} E(\boldsymbol{p}) &= \min_{t \ll \boldsymbol{p}} \{ D(t\|\boldsymbol{p}) + \tfrac{1}{2}[H(t) - 1]_+ \} \\ &= \max_{0 \le s \le 1} \frac{s}{1+s}[H_{1+s}(\boldsymbol{p}) - 1]. \end{aligned} \tag{14}$$

Here $D(t\|p) = \sum_i t_i \log_2(t_i/p_i)$ and $H_\alpha(p) = (1-\alpha)^{-1}\log_2 \sum_i p_i^\alpha$. The second line follows by writing $[H-1]_+/2 = \max_{0 \le u \le 1/2} u(H-1)$, exchanging extrema, and setting $s = u/(1-u)$. This is the known universal$_2$ privacy-amplification exponent at rate one [11], and Eq. (9) reproduces it without the polynomial prefactor. Applying (5) to empirical types t with $D(t\|\boldsymbol{p}) + H(t) < 1$ gives a matching lower bound near threshold. With $R_\mp$ the lim inf and lim sup of $-n^{-1}\log_2 \overline{T}_n$,

$$E(\boldsymbol{p}) \le R_- \le R_+ \le \inf_{D(t\|\boldsymbol{p}) + H(t) < 1} D(t\|\boldsymbol{p}). \tag{15}$$

For depolarizing noise both sides expand to the same leading order: $R_\pm = \kappa(r - r_c)^2 + O((r - r_c)^3)$ with $\kappa = 9/[32 \ln 2\, q_c(1 - q_c)] = 2.64\ldots$. Away from threshold the rate depends on the ensemble. Uniformly random stabilizer states can be written as $\psi = d^{-1}[I + \sum_{P \ne I} z_P P]$ with $z_P \in \{0, \pm 1\}$ and $\mathbb{E}|z_P| = 1/(d+1)$. Local depolarization multiplies a weight-w term by $(1-r)^w$, and the triangle inequality gives

$$\overline{T}_n^{\mathrm{stab}} \le \frac{[1 + 3(1-r)]^n - 1}{2^n + 1}. \tag{16}$$

At $r = 0.9$ this rate is at least $1 - \log_2(1.3) = 0.621\ldots$, larger than $E = 0.479\ldots$ from (14). Random stabilizer states form a 3-design [1], so matching three moments of Haar does not by itself make the achievable rate tight. This does not show that the Haar and stabilizer rates differ, and neither rate is determined here.

Discussion.—Encoding into a scrambled state does not make information fragile against local Pauli noise, provided nothing randomizes the noise afterward. The information survives up to the hashing entropy, where the error distribution has the same entropy as the Hilbert space. The bracket controlling this involves only the error distribution and the dimension, so it holds for every exact 2-design and for correlated noise. The remaining questions depend on more than two moments: the limit at degenerate critical noise, and decay rates away from threshold. For error mitigation, the bounds of Refs. [3, 4, 7] imply exponential cost at large depth for any noise strength. The present results say that after a single noisy layer the purity-based estimate is loose throughout $0.252 < r < 0.423$, because it ignores the low-surprisal errors that the optimal measurement exploits. Translating the average-case statement into worst-case error-mitigation cost, and extending it to approximate designs and non-Pauli channels, would need further arguments that we do not supply.

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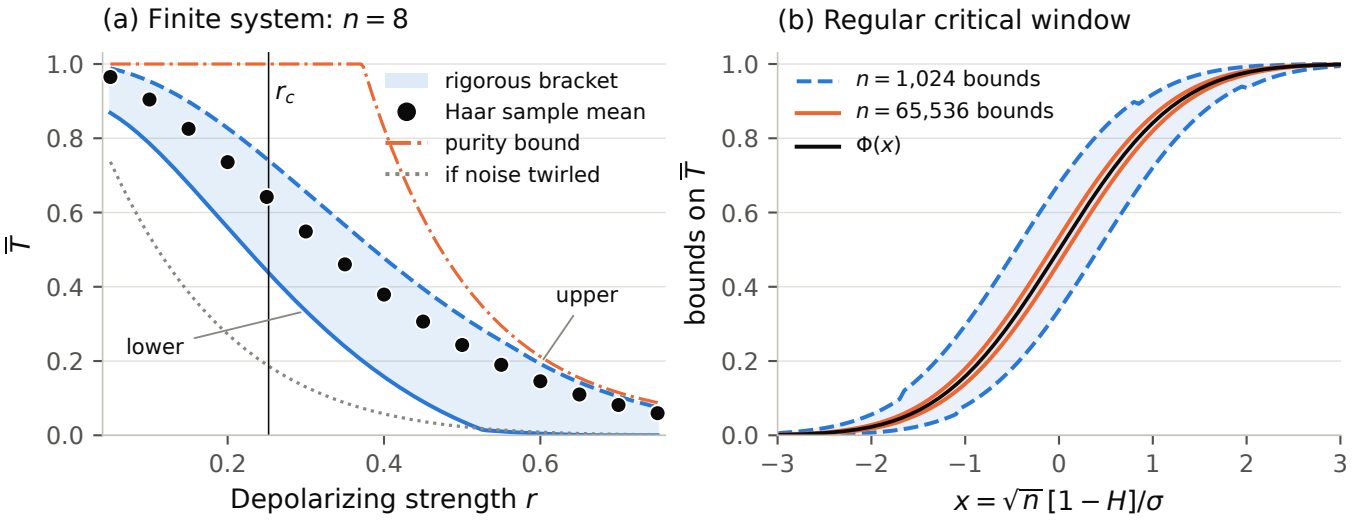


FIG. 1. Distinguishability after one layer of local depolarizing noise. (a) At $n = 8$, the shaded interval lies between the projector lower bound (solid) and the two-class upper bound (dashed) of Eqs. (5)–(6), optimized over error-weight cutoffs; the thin vertical line marks r_c. The lower bound vanishes for $r \geq 2/3$, where no single error has probability above $1/d$. Points are means over 64 independent Haar-state pairs, computed from full density matrices; bars are one sample standard error, and the same pairs are reused at every r. The dash-dotted curve is the single-class (purity) bound, which falls below one only for $r \gtrsim 0.37$ at this size and certifies decay only above 0.423 asymptotically. The dotted gray curve is the depolarizing parameter θ that would result if a further 2-design unitary twirled the noise; it equals the twirled distance up to a factor $1 - O(1/d)$. (b) The bracket (4) with $a = 2\log_2 n$, evaluated from binomial distributions at fixed $x = \sqrt{n}(1 - H)/\sigma$, for $n = 1024$ and 65536. The intervals approach $\Phi(x)$ (black). Panel (b) shows bounds, not simulations.

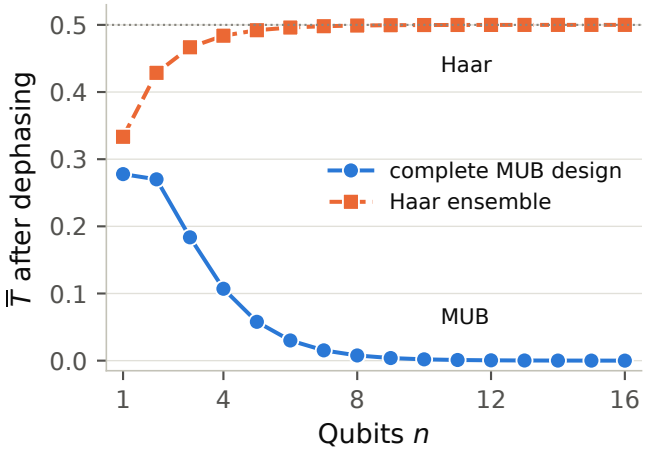


FIG. 2. Design dependence at a degenerate critical channel. Under complete dephasing both ensembles are exact 2-designs and the error entropy is one bit per qubit. Equations (12) and (13), evaluated without sampling, give different limits. The dotted gray line marks 1/2.

an OpenAI GPT model [version to be specified by the author] identified the omitted Ref. [8]. The author is responsible for all content. During preparation, proofs were rederived independently, the exact counterexamples were checked by explicit enumeration for $n \leq 2$, numerical claims were checked against independent implementations recorded with the data, and cited statements were checked against the primary sources.

Data availability.—The data that support the findings of this article and the code that generates every figure and numerical check accompany this submission and will be deposited in a public repository [22].

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